START FROM SCRATCH / HYPER-PACED LEARNING

Week 4: Build your first AI agent

You have built a chatbot. You have grounded it with RAG. Now you will build a system that can choose a next step, use a tool, observe the result, and know when to stop.

WEEK 2	WEEK 3	WEEK 4
CHATBOT	**RAG**	**AGENT**
The AI can talk	The AI can read trusted knowledge	The AI can act with safe tools

The shift

An agent is not simply a smarter chatbot. A chatbot responds. A RAG application retrieves relevant knowledge before responding. An agent can decide whether it needs information, choose a tool, use the result, and continue or escalate based on evidence.

RAG becomes a capability

Your Week 3 runbook search can now become one tool available to the agent. It is a reliable knowledge source - not automatically long-term memory.

Autonomy needs boundaries

The goal is not unrestricted action. The goal is useful, explainable action within clear limits and human approval points.

By the end of Week 4, your team should be able to explain why a task needs an agent, define its safe toolset, show the tool-use loop, and demonstrate a clear escalation path.

PREPARE BEFORE YOU BUILD

Three required resources

Complete these in order. Focus on the mental model and the decisions an agent must make - do not get lost in framework names or implementation details.

1 **Read: Building Effective AI Agents**
Anthropic
Read the sections that distinguish **workflows** from **agents**, explain when agents are useful, and introduce the building blocks. The important question: does this task need fixed steps, or does it need the model to choose the next step?
https://www.anthropic.com/engineering/building-effective-agents

2 **Watch: From Zero to Your First AI Agent in 25 Minutes**
YouTube
Use this as the visual bridge into agent building. Notice the goal, the available tools, the feedback from those tools, and the point at which the agent stops.
https://youtu.be/EH5jx5qPabU?si=Oh0ccE4oJWwSz7vi

3 **Skim: Function Calling in AI Agents**
Prompt Engineering Guide
Learn the handoff: the model proposes a tool call and its parameters; the application performs the action; the tool result returns as an observation for the model to use.
https://www.promptingguide.ai/agents/function-calling

Optional, after the core pre-work

Learn the loop: Hugging Face - What is an AI Agent? introduces Think -> Act -> Observe in beginner-friendly terms.

Keep for later: Skills vs MCP vs RAG vs Memory is useful once your first agent works. Do not let MCP or memory terminology distract from the basic tool-use loop this week.

The five agent types: an IBM lens

Simple reflex Reacts to the current input.	**Model-based** Uses an internal view of state.	**Goal-based** Selects actions to reach an outcome.	**Utility-based** Compares outcomes and trade-offs.	**Learning** Improves from feedback and experience.

This is a conceptual taxonomy. For Week 4, build a small **goal-based** agent that uses safe tools and feedback. Watch the optional IBM Technology overview for the visual explanation.

THE FIRST AGENT

Build a safe Triage and Resolution Agent

Continue the same SRE domain so the challenge is agent design, not learning a new business problem. Your agent should investigate an incident, retrieve evidence, recommend a path, and escalate when appropriate.

1. GOAL	2. DECIDE	3. USE TOOL	4. OBSERVE
Read the incident	Choose the next step	Retrieve or check	Use the result

Repeat only when the next tool call can improve the answer. Otherwise answer, escalate, or stop.

Start with three tools

Tool	What it does	Safety boundary
search_runbooks	Retrieves the most relevant Week 3 runbook guidance.	Read-only knowledge retrieval.
get_service_status	Checks a mocked or safe diagnostics/status source.	Read-only. Show the returned evidence.
draft_escalation	Prepares a concise escalation summary or ticket draft.	Draft only. A human approves submission.

Good first-agent boundaries

Use a small maximum number of tool calls. Display the tool calls and observations. Escalate when evidence is missing or conflicting. Never perform a live destructive action in the first build.

Avoid this week

No multi-agent architecture, autonomous retries, long-term memory, real ticket creation, real email sending, or framework-heavy abstraction. Prove one safe loop first.

COME READY TO DESIGN

Your team Agent Canvas

Complete this together before implementation. It will keep the agent grounded in a practical goal, a small toolset, and a credible safety model.

1. Goal What outcome should the agent achieve for the incident?	**2. Inputs** What information does it receive at the start?
3. Tools What can it retrieve, inspect, or draft?	**4. Observations** What does each tool return for the agent to use?
5. Decisions Which decision is the agent allowed to make?	**6. Stop condition** When must it answer, escalate, or stop trying?
7. Human gate Which action always needs approval?	**8. Demo proof** How will you visibly show the loop, evidence, and final outcome?

Bring these answers to the build

1. When would a fixed workflow be enough - and why does your use case need an agent?
2. Who actually executes a tool call: the model or the application?
3. What happens when the agent has insufficient or conflicting evidence?

Present the agent, receive feedback, and learn from every other team. Keep the problem statement clear, show more than you tell, and use the discussion to improve the next iteration.